Criterion 8: Continuous Improvement (80)

by
Dr. T V Rama Krishna
Professor of ECE

8. 1. Actions Taken Based on the Results of Evaluation of the COs, POs, and PSOs (40)

8.1.1. Actions Taken Based on the Results of Evaluation of the COs Attainment (20)

- **Evaluation of CO Attainment**
  - **Direct Methods:**
    - **Internal assessments, semester-end examinations, lab work, projects, viva, class performance.**
    - **Rubrics-based assessment of experiments, presentations, assignments.**
  - **Indirect Methods:**
    - **Student surveys, course exit surveys, alumni feedback, employer input.**
  - **Benchmarks/Thresholds:**
    - **Example: If 70% of students scoring ≥ 50 marks is the target, but only 55% achieved it, then a gap exists.**
  - **Gap Analysis:**
    - **Identify specific topics or skills where students underperformed.**
    - **Classify weaknesses as conceptual gaps, lack of practical exposure, or inadequate higher-order skill development.**
- **Areas of Weakness Identified (Examples)**
  1. **Low attainment in analytical/problem-solving oriented COs.**
  2. **Poor performance in application-based/laboratory COs.**
  3. **Weak fundamentals in mathematical and programming-related COs.**
  4. **Difficulty in higher Bloom's Taxonomy levels (Analyze, Evaluate, Create).**
  5. **Limited exposure to industry practices in project-related COs.**

- **Measures Taken / Improvement Actions**
  1. **Curriculum Intervention**
     - **Revision of syllabi to include additional topics or case studies.**
     - **Introduction of bridge courses in mathematics, coding, or communication.**
     - **Integration of emerging technology tools/software in labs.**
  2. **Pedagogical Initiatives**
     - **Use of ICT-enabled teaching (simulations, virtual labs, video lectures).**
     - **Flipped classroom approach and problem-based learning.**
     - **Assigning mini-projects, design tasks, and open-ended experiments.**
  3. **Support System Improvements**
     - **Remedial classes for slow learners.**
     - **Special tutorials, doubt-clearing sessions, peer-assisted learning.**
     - **Mentoring for academically weak students.**
  4. **Assessment Reforms**
     - **Redesigning question papers to balance Bloom's levels.**
     - **Rubrics for evaluating higher-order skills in labs/projects.**
     - **Frequent low-stakes quizzes to improve learning retention.**
  5. **Faculty Development**
     - **Training workshops on pedagogy, digital tools, and outcome-based teaching.**
     - **Interactions with industry experts for aligning teaching with real-world applications.**
  6. **Student Exposure and Motivation**
     - **Guest lectures, industrial visits, and internships.**
     - **Encouraging participation in hackathons, coding challenges, innovation contests.**
     - **Counseling and motivational sessions to reduce dropouts.**

- **Evidence and Documentation**
  - **CO Attainment Reports (course-wise, year-wise).**
  - **Action Taken Reports (ATR) of faculty meetings, DAB (Department Advisory Board) discussions.**
  - **Records of remedial classes, workshops, bridge courses, FDPs.**
  - **Improved attainment charts after actions were implemented.**
  - **Feedback forms from students and stakeholders confirming effectiveness.**
- **Impact Demonstration**
  - **Improved CO attainment in the next cycle (comparison of CAYm1 with previous years).**
  - **Increased pass percentages and reduction in failure rates.**
  - **Higher student satisfaction in feedback surveys.**
  - **Enhanced placement, internships, or project outcomes linked to improved COs.**

Sample Excel Layout

Course Code & Title	CO No.	Target Attainment (%)	Actual Attainment (%)	Attainment Gap (%)	Weakness Identified	Action Taken	Implementation Evidence	Impact/Improvement
CS301 – Data Structures	CO1	70	55	15	Poor understanding of recursion concepts	Conducted extra tutorial sessions & provided video lectures	Attendance sheets, shared PPTs, ATR of meeting	Improved to 68% in next cycle
CS301 – Data Structures	CO2	70	52	18	Lack of coding practice in lab	Introduced weekly coding challenges	Lab performance records, screenshots of online platform	Improved programming success rate by 20%
EE205 – Circuit Theory	CO3	65	50	15	Weak in problem analysis & circuit simulations	Added simulation-based assignments (MATLAB/PSPICE)	Assignment submissions, lab manuals	Attainment improved to 64%
ME210 – Thermodynamics	CO4	75	60	15	Conceptual clarity issues in entropy & laws of thermodynamics	Bridge course on basics + guest lecture by industry expert	Bridge course attendance, guest lecture report	Student feedback improved, attainment reached 72%

8.1.2. Actions Taken Based on the Results of Evaluation of the POs/PSOs Attainment (20)

Evaluation Process of POs/PSOs

- **Direct Assessment:**
  - **Aggregated CO attainment mapped to relevant POs/PSOs.**
  - **Capstone projects, internships, labs, and design-oriented activities.**
- **Indirect Assessment:**
  - **Exit surveys, alumni feedback, employer surveys, placement statistics, higher studies.**
- **Benchmarking & Gap Analysis:**
  - **Targets set (e.g., 70%).**
  - **Actual attainment calculated.**
  - **Identify gaps (e.g., PO4 “Conducting Investigations” attained at 55% vs. target 70%).**

Areas of Weakness Typically Identified

1. **Technical/Problem-Solving Skills (PO2, PO3, PO4, PO5):**
   - **Students struggle with analytical/problem-solving tasks.**
2. **Research Orientation & Lifelong Learning (PO4, PO12):**
   - **Limited exposure to open-ended investigations or self-directed learning.**
3. **Communication & Teamwork (PO9, PO10):**
   - **Weak presentation, report writing, and team collaboration skills.**
4. **Ethics & Professional Responsibility (PO8):**
   - **Need reinforcement of professional ethics and sustainability practices.**
5. **Program-Specific Outcomes (PSOs):**
   - **Lack of industry-grade software/tools.**
   - **Limited exposure to latest domain technologies.**

Measures Identified and Implemented (Two Years)

1. **Curriculum Interventions**
   - **Introduced electives on emerging technologies (AI/ML, IoT, Sustainability).**
   - **Revision of lab manuals to include industry-oriented experiments.**
   - **Integration of mini-projects & case studies into courses.**
   - **Inclusion of multidisciplinary and open electives.**
2. **Pedagogical Initiatives**
   - **Shift from lecture-based to active learning (flipped classroom, problem-based learning).**
   - **Adoption of ICT tools (Moodle, virtual labs, coding platforms, simulation software).**
   - **Rubrics-based assessment to measure teamwork, ethics, and communication.**
   - **Guest lectures and expert talks linked to specific POs/PSOs.**

5. Support System Improvements

- **Remedial classes and bridge courses for weak students.**
- **Peer-assisted learning groups and mentoring system.**
- **Soft-skills and communication training modules.**
- **Career guidance and placement support for industry-readiness.**

6. Faculty Development

- **FDPs on Outcome-Based Education and new pedagogies.**
- **Industry training for faculty to align teaching with real-world practices.**
- **Internal workshops on assessment of higher-level POs (ethics, lifelong learning).**

7. Industry & Research Linkages

- **MoUs with industries for internships and live projects.**
- **Hackathons, ideation camps, and innovation challenges aligned with POs.**
- **Encouraging student research publications and conference participation.**

Documentation and Evidence

- PO/PSO Attainment Analysis Reports (year-wise).
- Action Taken Reports (ATR) from Department Advisory Board (DAB), BoS, Academic Council.
- Evidence of interventions:
  - Remedial/bridge course attendance.
  - FDP/workshop circulars & reports.
  - Internship/project completion reports.
  - Student/alumni/employer feedback forms.
- Improvement Charts: PO/PSO attainment before vs. after interventions (CAYm2 → CAYm1 → CAY).

Impact & Continuous Improvement

- Demonstrated Improvements:
  - Example: PO4 (Investigation of problems) improved from 55% → 68% over two years due to simulation labs & projects.
  - PO9 (Communication) improved due to seminar courses & presentation rubrics.
  - PSO1 (Domain-specific tools) improved after industry tie-ups & new lab installations.
- Closing the Loop:
  - Actions → Implementation → Evidence → Measurable improvement → Reviewed in next cycle.

Format of Year-wise PO/PSO Attainment Report

PO/PSO Code	Target Attainment (%)	CAY (2024-25)	CAYm1 (2023-24)	CAYm2 (2022-23)	Weakness Identified	Actions Taken	Impact/Improvement
PO4 – Investigation	70	68	61	55	Weak in analytical & investigation skills	Simulation-based labs, mini-projects	Improved from 55% → 68%
PO9 – Communication	75	72	66	60	Poor presentation & writing skills	Seminar course, rubrics, peer reviews	Communication attainment ↑12%
PO12 – Lifelong Learning	70	71	63	58	Lack of self-learning initiatives	MOOC credits, NPTEL/Coursera	80% students completed MOOCs
PSO1 – Domain Tools	70	66	60	52	Limited exposure to industry tools	Installed software, training sessions	Attainment gap reduced by 14%

8.2. Academic Audit and Actions Taken thereof during the Period of Assessment (15)

Components of Academic Audit

1. **Course Delivery Audit**
   - **Syllabus coverage, lesson plans, teaching methodologies.**
   - **Alignment of COs with curriculum delivery.**
2. **Assessment Audit**
   - **Quality of internal tests, assignments, lab records.**
   - **Bloom's Taxonomy compliance in question papers.**
   - **Fairness in evaluation and moderation process.**
3. **Outcome Attainment Audit**
   - **Review of CO, PO, PSO attainment levels.**
   - **Validation of direct & indirect assessment methods.**
4. **Research & Innovation Audit**
   - **Student/faculty publications, projects, patents, industry collaborations.**
5. **Faculty Performance Audit**
   - **Teaching effectiveness, FDP participation, research contributions.**
6. **Student Support Audit**
   - **Mentoring, remedial classes, bridge courses, internships, placements.**

Process of Academic Audit

- **Step 1: Planning**
  - **Academic Council / IQAC prepares academic audit framework.**
- **Step 2: Data Collection**
  - **Department submits course files, CO–PO mappings, attainment reports, and feedback data.**
- **Step 3: Audit Committee Review**
  - **Internal committee (HoD, senior faculty, IQAC) and sometimes external experts review processes.**
- **Step 4: Identification of Strengths & Weaknesses**
  - **Example: *"Strength: Strong industry internships; Weakness: Low attainment in PO4 (Investigations)."***
- **Step 5: Recommendations & Action Plan**
  - **Corrective measures proposed for low attainment areas.**
- **Step 6: Implementation & Follow-up**
  - **Departments implement actions and present Action Taken Reports (ATR) in the next cycle.**

Examples of Actions Taken

1. **Curriculum & Pedagogy**
   - **Revised course content to include industry-relevant modules.**
   - **Adopted flipped classroom and ICT-based teaching.**
2. **Assessment Reforms**
   - **Introduced rubrics for project evaluation.**
   - **Improved question paper design to test higher-order skills.**
3. **Faculty Development**
   - **FDPs on outcome-based education and pedagogy.**
   - **Training on domain-specific tools/software.**
4. **Student Support Enhancements**
   - **Remedial classes for slow learners.**
   - **Peer-assisted learning and mentoring programs.**
   - **Bridge courses in math/programming basics.**
5. **Infrastructure & Industry Linkages**
   - **Lab upgradation with modern simulation tools.**
   - **MoUs signed for internships and live projects.**

Documentation & Evidence

- Academic Audit Reports (year-wise).
- Minutes of Academic Audit Meetings with recommendations.
- Action Taken Reports (ATR) on audit findings.
- Improved attainment charts after corrective measures.
- Feedback forms from students, alumni, employers.

Impact of Academic Audits

- Progressive improvement in attainment levels across COs/POs/PSOs.
- Better curriculum delivery and assessment practices.
- Improved student performance, placements, and higher studies.
- Enhanced faculty competency through FDPs.
- Increased stakeholder satisfaction (students, parents, employers, alumni).

Marking Scheme (15 Marks)

- Audit system & process clarity – 5 Marks.
- Implementation with evidence – 5 Marks.
- Corrective actions & measurable improvements – 5 Marks.

Suggested Year-wise Table Format

Academic Year	Criterion Audited	Strengths Identified	Gaps/Weaknesses	Recommendations	Action Taken	Evidence	Impact
2022-23 (CAYm2)	Course Delivery	Syllabus coverage > 95%	Low attainment in CO2	Introduce coding workshops	Weekly coding practice sessions introduced	Audit report, attendance	CO2 attainment improved by 12%
2023-24 (CAYm1)	Assessment	Moderate question paper quality	Few higher-order Bloom's questions	Train faculty in assessment design	FDP on question paper setting conducted	FDP certificates	Increase in higher-order questions to 40%
2024-25 (CAY)	Student Support	Strong mentoring system	Weak in lifelong learning (PO12)	Encourage MOOCs	MOOCs made mandatory with credits	Certificates uploaded	PO12 improved from 63% → 71%

8.3. Improvement in Faculty Qualification/Contribution (10)

Areas Considered for Assessment

A. Improvement in Qualification

- **Faculty pursuing and completing higher degrees:**
  - **M.E./M.Tech → Ph.D.**
  - **Ph.D. → Post-Doctoral research.**
- **Faculty completing professional certifications (NPTEL, Coursera, AICTE ATAL FDPs, domain-specific certifications).**

B. Research & Publications

- **Increase in journal/conference publications (Scopus, SCI, WoS indexed).**
- **Book chapters, patents filed/granted, copyrights.**
- **Participation and presentations in national/international conferences.**

C. Funded Research & Consultancy

- **Grants received from AICTE, DST, UGC, industries.**
- **Industry consultancy projects undertaken.**

D. Faculty Development & Contributions

- **FDPs, STTPs, MOOCs completed or organized.**
- **Guiding UG/PG/Ph.D. students.**
- **Memberships in professional bodies (IEEE, ISTE, ACM, ASME).**
- **Contribution to curriculum design, academic audits, NBA/NAAC activities.**

Process of Evaluation

- **Step 1: Baseline Data (CAYm2) – Faculty qualifications, publications, research at the start of assessment period.**
- **Step 2: Year-wise Tracking (CAYm1, CAY) – Record newly acquired qualifications, increased publications, FDPs attended, research projects sanctioned.**
- **Step 3: Comparative Analysis – Improvement measured as growth % or numbers.**
- **Step 4: Evidence Collection – Degrees awarded, publication proofs, sanction letters, FDP/Workshop certificates.**

Suggested Year-wise Analysis Table

Year	Faculty Strength	Ph.D. Holders	Faculty Enrolled in Ph.D.	Journal Publications	Conference Publications	FDPs/STTPs Attended	Patents Filed/Granted
CAYm2 (2022-23)	28	12	5	18	10	15	2/0
CAYm1 (2023-24)	30	14	4	25	15	22	3/1
CAY (2024-25)	31	17	3	32	20	30	4/2

8.4. Improvement in Academic Performance (15)

1. Purpose

- To measure year-wise growth in student academic performance across 1st, 2nd, and 3rd years during the assessment period (CAY, CAYm1, CAYm2).
- Provides evidence of how institutional teaching-learning processes, remedial measures, mentoring, and academic support have contributed to better outcomes.
- Ensures alignment with Outcome-Based Education (OBE) and continuous quality improvement.

2. Parameters Considered

NBA typically evaluates using Academic Performance Index (API) and progression trends:

1. Pass Percentage – Ratio of students passed to students appeared in each year.
2. Average Academic Performance Index (API)
3. Year-wise Improvement Trends – Performance compared for CAY, CAYm1, CAYm2.
4. Comparison of Weak vs. Strong Students – Tracking impact of remedial classes, bridge courses, and mentoring.

Sample Data Presentation

Year	Academic Year	No. of Students Appeared	No. of Students Passed	Pass %	Avg. GPA/10	API	Observation/Improvement
1st Year	CAYm2 (2022-23)	120	95	79%	6.8	0.54	Baseline year – gaps in mathematics & programming
1st Year	CAYm1 (2023-24)	125	108	86%	7.2	0.62	Improvement due to bridge courses & remedial sessions
1st Year	CAY (2024-25)	130	120	92%	7.8	0.71	Strong improvement with mentoring + ICT tools

Corrective Measures that Drive Improvement

- Bridge Courses → Mathematics, Programming, Communication.
- Remedial Classes → For academically weak students.
- Mentoring System → Faculty advisors + peer mentoring.
- Pedagogical Innovations → ICT-enabled classrooms, flipped learning, simulation tools.
- Workshops/Internships → Industry exposure, mini-projects, hackathons.
- Faculty FDPs → Trained faculty deliver more effective teaching.

Evidence Required

- Year-wise exam results & grade analysis.
- API calculations (CAY, CAYm1, CAYm2).
- Records of remedial/bridge courses.
- Minutes of meetings / ATRs linking academic gaps with actions.
- Improvement charts (bar/line graphs for pass % and API).